Customer Segmentation Using Unsupervised Learning

Final project report based on the Telco Customer Churn dataset

Executive Summary

This analysis used unsupervised learning to identify distinct customer segments in the Telco Customer Churn dataset. Three clustering approaches were compared: K-Means, hierarchical clustering, and DBSCAN. K-Means with three clusters was selected as the final model because it delivered the strongest silhouette score among the tested variants and produced segments that were easier to interpret and translate into business action. The results show clear differences in tenure, spending, contract structure, and churn exposure across customer groups.

Main Objective

The objective of this analysis is to identify meaningful customer segments using unsupervised learning techniques, in order to support more targeted retention, pricing, and service strategies. This is a clustering project rather than a supervised prediction task. The goal is to uncover hidden structure in the customer base and translate those patterns into decisions that matter to stakeholders.

Data Description

The dataset contains 7,043 customer records and 21 variables. It includes demographic, billing, and service-related attributes such as tenure, MonthlyCharges, TotalCharges, Contract, InternetService, and PaymentMethod. The original source is the widely used Telco Customer Churn dataset. Although the dataset also includes a Churn field, that variable was not used to create clusters. Instead, it was retained for post hoc interpretation to assess whether certain segments also showed higher churn exposure.

Data Exploration and Preparation

Data quality checks showed that TotalCharges contained non-numeric values caused by blank entries. These were converted to numeric with coercion, and rows with missing TotalCharges were removed for the clustering workflow. For a clear and

computationally efficient segmentation baseline, the final feature set used for clustering was limited to four numeric variables: tenure, MonthlyCharges, TotalCharges, and SeniorCitizen. These features were standardised before model fitting so that distance-based methods did not become dominated by larger-scale variables such as total spend.

Model Variations and Comparative Evaluation

Three unsupervised learning approaches were compared on a representative sample of the standardised feature space: K-Means, hierarchical clustering with Ward linkage, and DBSCAN. K-Means was tested with 3, 4, and 5 clusters. Hierarchical clustering was tested with 3 and 4 clusters. DBSCAN was tested with eps values of 0.5 and 0.7. Silhouette score was used as the primary comparison metric because it captures how well each observation fits within its assigned cluster relative to neighbouring clusters.

Model variation	Silhouette score	Comment
K-Means (k=3)	0.464	3 centroid-based clusters
K-Means (k=4)	0.427	4 centroid-based clusters
K-Means (k=5)	0.452	5 centroid-based clusters
Hierarchical (Ward, k=3)	0.431	Dendrogram-style hierarchical merge
Hierarchical (Ward, k=4)	0.36	Dendrogram-style hierarchical merge
DBSCAN (eps=0.5)	0.361	Noise-aware density clustering; noise points=64
DBSCAN (eps=0.7)	0.415	Noise-aware density clustering; noise points=16

K-Means with three clusters achieved the highest silhouette score among the tested variants (0.464), ahead of hierarchical clustering with three clusters (0.431) and the strongest DBSCAN setting tested (0.415). DBSCAN also produced noise points and less balanced, less interpretable groupings. As a result, K-Means with three clusters was selected as the final model.

Final Model Recommendation

The recommended model is K-Means with three clusters. It best fits the project objective because it offers a useful balance of analytical clarity, interpretability, and

stakeholder value. The resulting segments are distinct enough to support action, simple enough to explain to a senior audience, and more stable than the denser but noisier DBSCAN output.

Key Findings and Insights

The final model identified three broad customer segments with different commercial profiles. Cluster summaries are shown below.

Cluster	Segment	Customers	Avg tenure	Avg monthly	Avg total	% senior	Churn rate	Typical contract	Typical payment
0	Established premium customers	1875	57.24	88.63	5056.32	0.0	0.14	Two year	Credit card (automatic)
1	Lower-spend newer customers	4015	20.58	49.4	838.36	0.0	0.28	Month-to-month	Mailed check
2	Higher-risk senior customers	1142	33.3	79.82	2810.47	1.0	0.42	Month-to-month	Electronic check

Cluster 0 (Established premium customers): 1875 customers. Average tenure 57.24, average monthly charges 88.63, and churn rate 0.14. The most common contract type is Two year. This segment can be summarised as a distinct commercial profile rather than a random grouping.

Cluster 1 (Lower-spend newer customers): 4015 customers. Average tenure 20.58, average monthly charges 49.4, and churn rate 0.28. The most common contract type is Month-to-month. This segment can be summarised as a distinct commercial profile rather than a random grouping.

Cluster 2 (Higher-risk senior customers): 1142 customers. Average tenure 33.3, average monthly charges 79.82, and churn rate 0.42. The most common contract type

is Month-to-month. This segment can be summarised as a distinct commercial profile rather than a random grouping.

The most commercially important pattern is that the higher-risk senior segment shows the highest churn rate (0.42), while established premium customers have the lowest churn exposure (0.14). The lower-spend newer segment is much larger and sits in the middle on churn risk (0.28), making it a practical target for onboarding and retention interventions.

For stakeholders, the implication is straightforward: do not treat the customer base as one homogeneous population. Retention and service strategies should be segment-specific. Senior high-risk customers need proactive support and simpler retention offers. Newer lower-spend customers need better early-life engagement. Established premium customers are valuable and comparatively stable, so the priority there is protection and loyalty rather than aggressive intervention.

Visualisation

The PCA chart below provides a two-dimensional view of the final K-Means segmentation. It is not the basis for model selection, but it helps communicate the presence of broad segment separation.

Limitations

This analysis has real limitations. First, the clustering model used only four numeric variables to keep the workflow interpretable and computationally efficient. That choice improves clarity but ignores potentially useful categorical information such as service bundles and payment behaviour during cluster formation. Second, cluster quality depends on distance assumptions and scaling choices. Third, unsupervised clusters do not come with a ground-truth label, so usefulness must be judged by separation, interpretability, and business relevance rather than raw predictive accuracy. Finally, some overlap remains between segments, which is normal in customer data and means these clusters should be treated as decision support, not absolute truth.

Next Steps

The next iteration should expand the feature set to include encoded categorical variables such as Contract, InternetService, and PaymentMethod, then re-test K-Means, hierarchical clustering, and model-based approaches such as Gaussian

Mixture Models. Additional validation should include stability checks across random samples, richer segmentation metrics, and downstream testing against business outcomes such as revenue retention or support usage. A practical commercial next step would be to profile campaign response by cluster and assess whether segment-specific interventions outperform a one-size-fits-all strategy.